

Where the lock-in is now

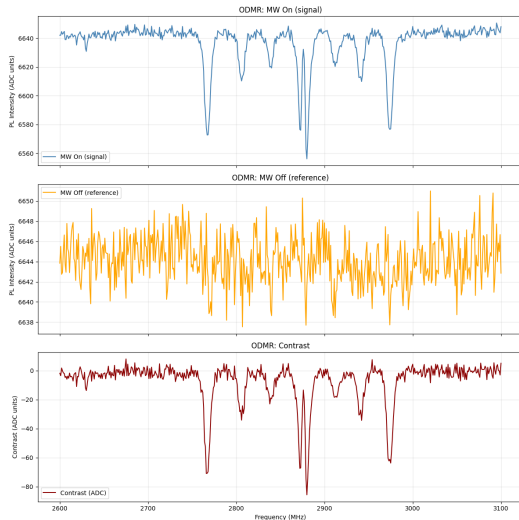
What's working

- ▶ Full pipeline runs end-to-end: ODMR sweep → parked-frequency plan → multipoint acquire → Δf inference → B-vector reconstruction.
- ▶ All 16 parked frequencies measured in a single FPGA program (MultipointLockinODMR) — no more per-frequency upload overhead.
- ▶ Reference shot is now genuinely MW-off: a long ($\sim 1700 \mu\text{s}$) relax delay between signal and reference lets the spins repolarize, so the reference no longer carries dip structure.
- ▶ Live mode builds the program once outside the batch loop and uses `pre_init=False` — the per-batch peaking spike from `01_basic` is gone.

May 7 run snapshot

ODMR source	20260507_140754
Parked points	16
Transitions	8
Bias field	-2.30, -2.99, -3.14 mT
Single-batch acquire	116 ms
Live run	281 samples, 21.0 s
Live rate	13.4 Hz
Vector rank-ok	281 / 281

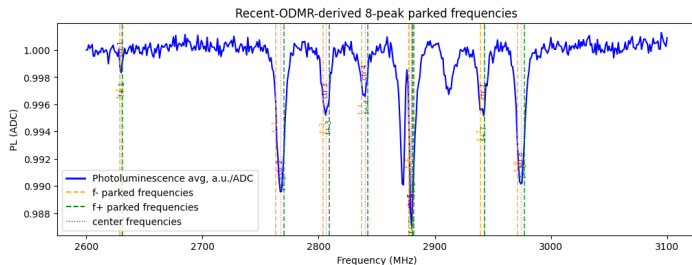
Reference shots are actually MW-off now



What changed

- ▶ Each sweep point now grabs MW-on signal and MW-off reference in one pass.
- ▶ Long ($\sim 1700 \mu\text{s}$) delay between the two integrations lets the always-on laser repolarize the spins, so the reference doesn't carry residual dip structure.
- ▶ The contrast trace = signal - reference is what feeds parked-frequency planning.

Parked-frequency plan from the ODMR fit



Plan output

Blk	Center MHz	Line MHz	f_- MHz	f_+ MHz
1	2630.0	4.0	2629.0	2632.0
2	2767.4	9.9	2764.0	2770.0
3	2806.0	4.0	2803.0	2809.0
4	2840.0	4.0	2837.0	2843.0
5	2880.6	3.4	2878.0	2882.0
6	2880.6	3.6	2878.0	2882.0
7	2941.5	4.5	2939.0	2944.0
8	2974.9	10.5	2971.0	2978.0

Custom program

The 16 points aren't evenly spaced — they're the steepest-slope flank points of the 8 NV transitions. A stock QICK linear sweep can't hit non-uniform frequencies, hence MultipointLockinODMR on the next slide.

One FPGA program for all 16 parked points

What it does

- ▶ One MultipointLockinODMR program holds the ASM for all 16 frequencies — uses `mw_frequency_register.set_to(f)` inside `body()` to hop between them.
- ▶ Predicted FPGA work: 68 ms per batch ($6.8 \text{ ms/rep} \times 10 \text{ reps}$). Measured first batch: 116 ms.
- ▶ Output: `multipoint_lockin_collected.csv` — one wide row per batch with `peak_NN`, `peak_NN_ref`, `peak_NN_freq_mhz`.

Build once, not per batch

If you instantiate `MultipointLockinODMR` inside the batch loop, it re-uploads the FPGA program every iteration ($\sim 300 \text{ ms}$ each — kills the speedup). Build once outside, then call `prog.acquire(progress=False)` in the loop.

Example columns from the wide CSV

Peak	Freq MHz	Signal ADC	Ref ADC
01	2629.0	6529.6	6535.3
02	2632.0	6539.4	6543.4
09	2878.0	6502.5	6507.1
10	2882.0	6488.6	6498.9
15	2971.0	6499.1	6509.3
16	2978.0	6499.2	6510.1

Speedup vs the old loop

Old 17-program loop at `reps=100`: $\sim 6.6 \text{ s}$ per batch, worse with network jitter. New multipoint at the same `reps`: $\sim 700 \text{ ms}$ per batch after the one-time upload — about $10\times$ faster.

Two-Point Δf Formula

Derivation

Linearize the spectrum at each parked point as $f_0 \rightarrow f_0 + \Delta f$:

$$S_-(t) \approx b_- - m_- \Delta f, \quad S_+(t) \approx b_+ - m_+ \Delta f.$$

Subtract to cancel common-mode drift:

$$\Delta d_{\text{now}} - \Delta d_{\text{ref}} = (m_- - m_+) \Delta f.$$

$$\Delta f_k = \frac{\Delta d_{\text{now}} - \Delta d_{\text{ref}}}{m_- - m_+}, \quad f_{\text{new},k} = f_{0,k} + \Delta f_k$$

Symbols

$b_{\pm}$, $m_{\pm}$: PL and slope at $f_{\pm}$ from the reference ODMR.

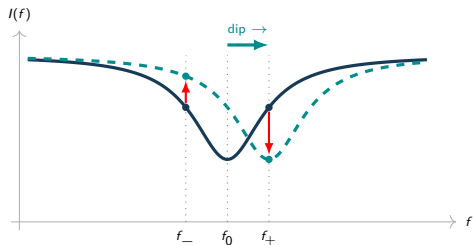
$$\Delta d_{\text{now}} = S_+(t) - S_-(t); \quad \Delta d_{\text{ref}} = b_+ - b_-.$$

$$\Delta B_{\text{proj},k} = \Delta f_k / \|df/dB\|_k.$$

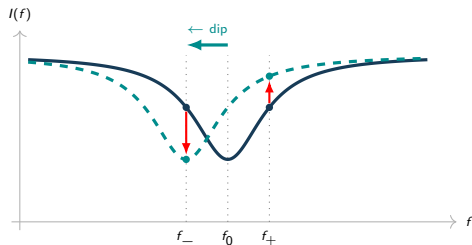
Block 4, latest batch

m_-, m_+	$-4.17 \times 10^{-4}, +5.36 \times 10^{-4} / \text{MHz}$
$\Delta d_{\text{now}} - \Delta d_{\text{ref}}$	-9.43×10^{-4}
Δf	$+0.990 \text{ MHz}$
f_{new}	2862.979 MHz

Why Denominator is $m_- - m_+$, Not $m_+ - m_-$



(a) $\Delta f > 0$: $\Delta d_{\text{now}} - \Delta d_{\text{ref}} < 0$

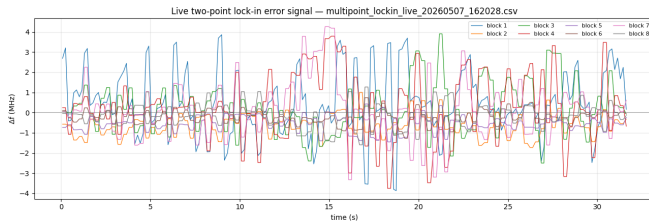


(b) $\Delta f < 0$: $\Delta d_{\text{now}} - \Delta d_{\text{ref}} > 0$

Sign self-consistency

Slope signs are fixed by the dip shape: $m_- < 0$ (left flank descends), $m_+ > 0$ (right flank ascends), so $m_- - m_+ < 0$ *always*. The numerator flips sign with the dip's direction (panels a, b above). Combining: $(-)/(-) = +$ in (a) and $(+)/(-) = -$ in (b), so Δf inherits the dip's true sign of motion. Flipping the denominator order to $m_+ - m_-$ (always > 0) would invert every Δf .

Live run: drift and noise floor



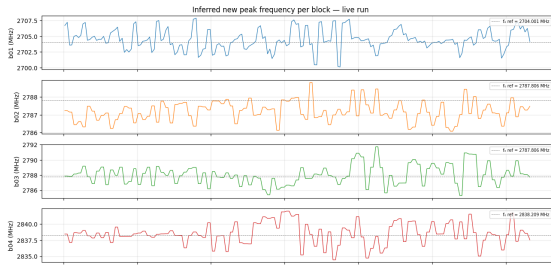
Run summary

- ▶ Live CSV 20260507_141355.
- ▶ 281 batches over 21.0 s $\rightarrow$ 13.4 Hz update rate.
- ▶ Per-batch timing: predicted 68 ms; actual median 12 ms, p95 15 ms, max 17 ms.
- ▶ Watchdog never fired.

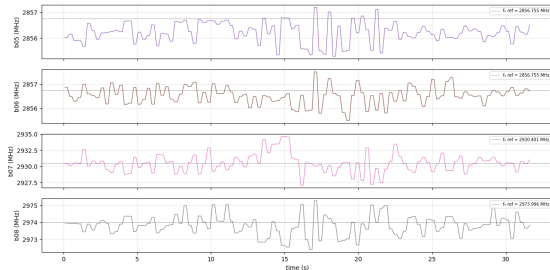
Per-block stats

Blk	mean Δf MHz	std Δf MHz	std ΔB uT
1	-0.211	0.354	12.6
2	-0.062	0.172	6.1
3	-0.583	0.460	16.3
4	+0.973	0.619	22.3
5	-0.645	0.102	7.4
6	-0.723	0.099	7.2
7	+0.024	0.279	9.9
8	-0.201	0.128	4.6

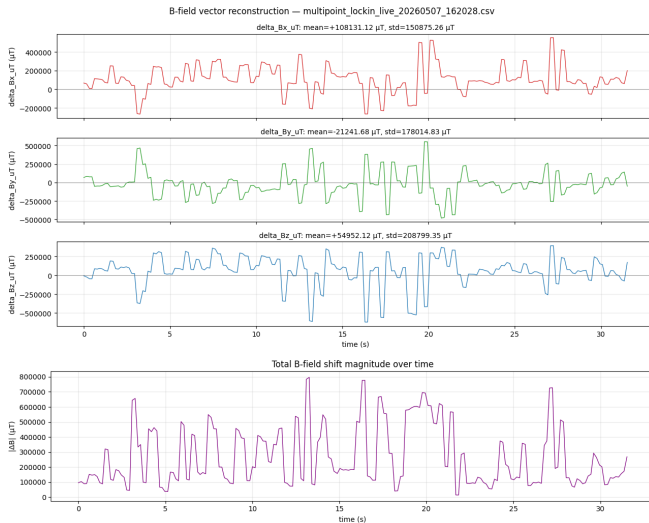
Inferred peak frequencies (live)



Each block: $f_{\text{new}}(t) = f_0 + \Delta f(t)$ in solid, with the dashed reference f_0 from the ODMR calibration.



B-field reconstruction output



Reconstruction stats

- ▶ Input: wide live CSV from the 281-sample run above.
- ▶ Samples: 281; rank-ok: 281/281.
- ▶ Matrix condition number: 14.08.

Vector component stats

Comp.	Mean uT	Std uT	Range uT
ΔB_x	18,923	81,622	-351,633 to 248,403
ΔB_y	-290,071	221,862	-942,851 to 688,841
ΔB_z	79,653	104,794	-323,599 to 634,124

What's next

ΔB magnitudes are huge — likely the bias hasn't been subtracted yet, or the inversion needs a sanity check against a known applied field. Also want to replot peak_01 vs acq_seconds from the CSV (outside the live cell) to confirm there's no batch-edge timing spike hiding in the data.